

Energy Balancing for Revenue Protection - TGI Use Case Notes



[UC28-01] - Awesense Energy Transition Platform Use Case

Overview

This manual is intended to:

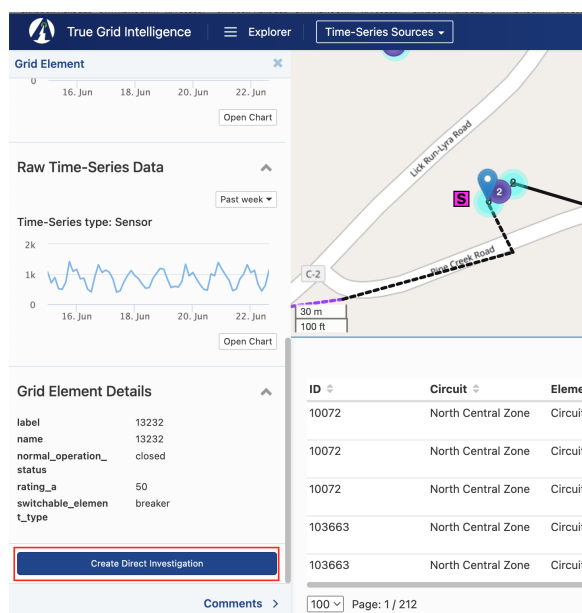
- Provide instructions to create the selected use case in True Grid Intelligence (TGI).

For more details about Energy Balancing for Revenue Protection and how it can be analyzed using Awesense's Energy Transition Platform, please refer to the [UC28-01 - Energy Balancing for Revenue Protection.pdf](#) document.

TGI Use Case Notes

This document shows the implementation process of mentioned use case in the TGI visualization interface along with a description of how to create such a use case on your data.

The usual way of getting the energy balance in TGI is to create an investigation. This can be created, for example, in Explorer (select from the main menu - top hamburger icon)¹. A measurement sensor with measured data must be found and a Placement must be created at the location of this sensor. The measuring sensor is selected from the map or table and click on the "Create Direct Investigation" button in the left side table at the bottom.



¹ There are other ways of creating investigation in TGI like for example:

- Placement Page and then creating direct investigation;
- Segment Analyzer Page and then creating investigation from segment;
- Investigation Page and then creating sub-investigation.

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In the Investigation, the start of the investigation has to be set. To calculate the energy balance, it is necessary to select for which segment we want to calculate the balance. In this case, it is the Target Segment. So in the left sidebar, we click on the "Target Segment" button and then on the "Recalculate Energy Balance" button.

True Grid Intelligence | Investigations » Investigation of 39660 | Map

Circuit
North Central Zone

GIS Date
1 Jan, 2021 9:00 PM

Owner
[awesense-petr](#)

Starts
1 Jun, 2023 5:23 PM

Description
No Description
[Show more / Update](#)

Type
Exploratory

State
Planning

Recoverability
Unknown

Ends
☒ Ongoing

Target Area

Transformers | Meters | Consumers | Producers (DERs)

Transformers	A	Σ
Count	56	56
kVA	948	948

[Zoom to Investigation](#) | [Consumer Breakdown](#)

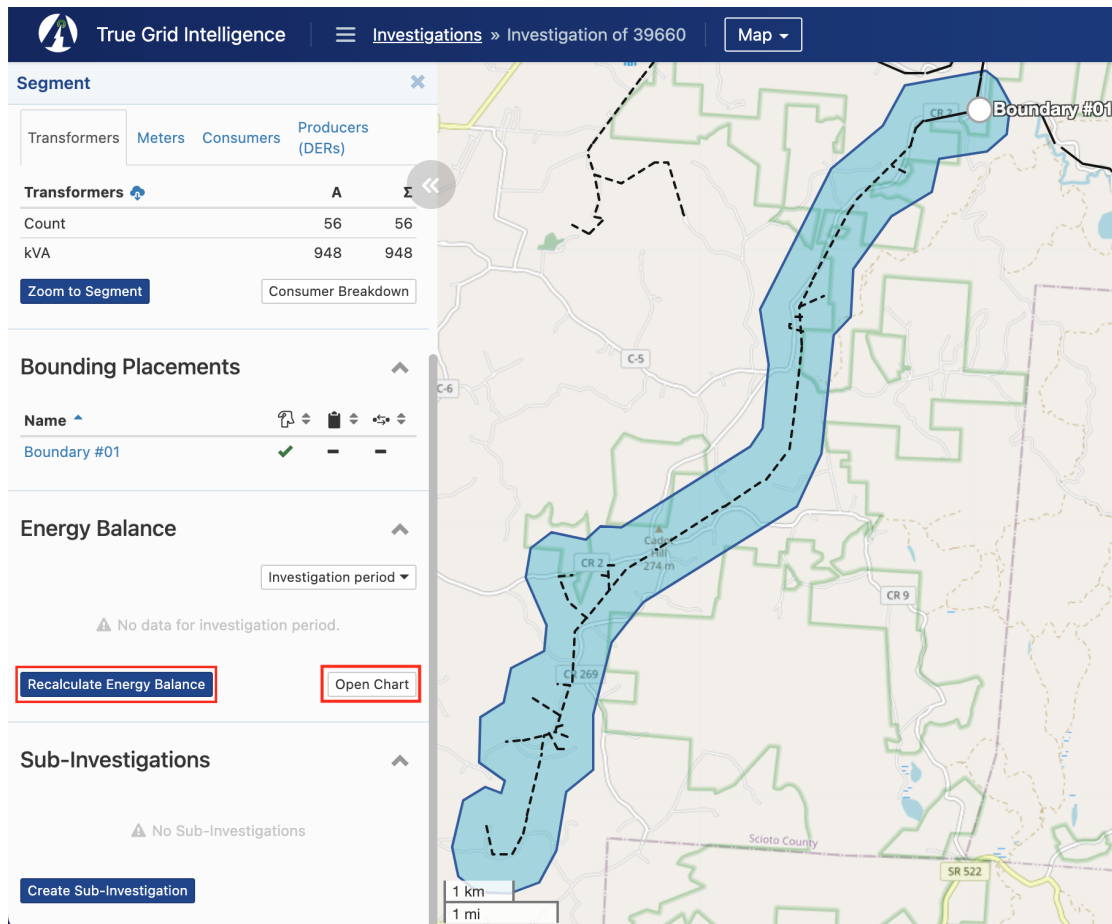
Segments

Name		
Target Segment	X	-

Map view showing a blue highlighted area labeled "Boundary #01" and "CR 2". The map includes labels for "C-5", "C-6", "CR 2", "CR 9", "Scioto County", and "SR 522". A scale bar indicates 1 km and 1 mi.

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Once calculated, the Energy Balance is displayed. To explore this balance in more detail, click on the "Open Chart" button.

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Energy Balance

Discrepancy: 61,293 kWh (46.7%), 3,284 kWh/day (18.7 days)

Recalculated on 22 Jun, 2023 5:34 PM

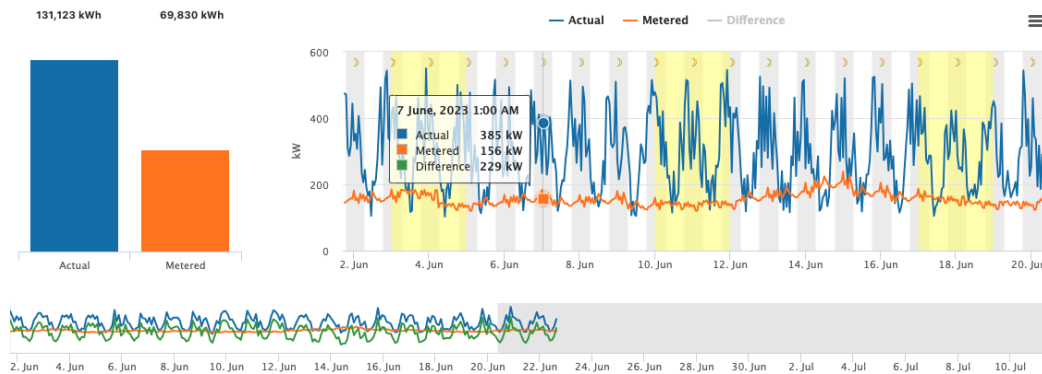
Recalculate Energy Balance

1 Jun, 2023 - 20 Jun, 2023

Show More Options

ACTUAL VS METERED

PROFILE



The energy balance graph shows the energy difference between the measured points and real consumption. The basic interface of the energy balance chart offers the possibility to change the different time periods of the energy balance. To quickly assess the state, summary information is written for the total discrepancy and losses per day.

Using the "Show More Options" button it is possible to display other supporting indicators such as Show Temperature, Show Weekends, Show Nights etc. for easier analysis of the values.

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Energy Balance

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Recalculate Energy Balance

1 Jun, 2023 - 20 Jun, 2023

Show Less Options

Units
kWh

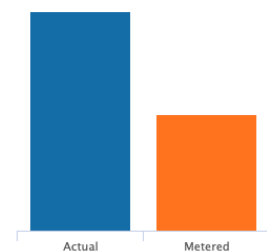
Chart Type
Line Chart

☒ Show Nights
☒ Show Weekends
☐ Show Temperature

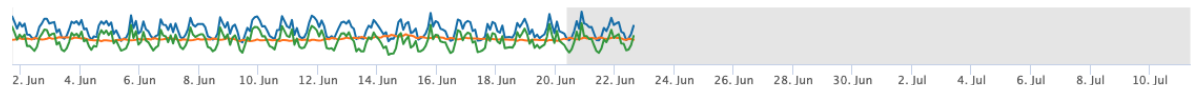
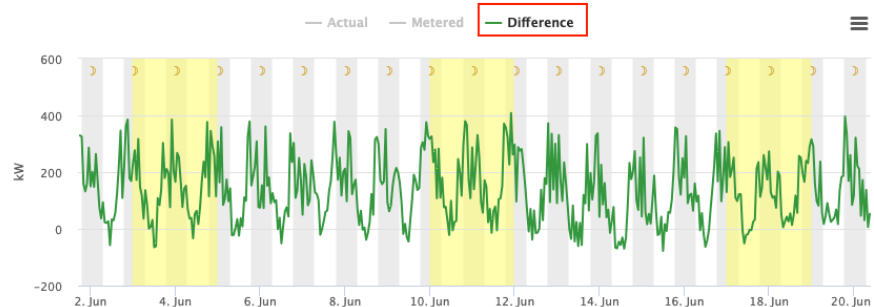
ACTUAL VS METERED

131,123 kWh

69,830 kWh



PROFILE



Using the "Difference" button, it is possible to display the difference of these measured energies over time and simply analyze in which hours or days the unauthorized consumption occurs. Using this feature, some signature patterns can be identified and the source of losses can be revealed more easily.

General Note

- This description was created for TGI version 3.10.0. For future versions of TGI, the process might be slightly different.
- Access to the different parts of TGI varies depending on the permissions of the user.
- Users on the Sandbox environments currently have read-only permissions, therefore, this use case may be accessible only for viewing, not creation or editing. If you are interested in the latter, please contact Awesense.
- Some of the data for this use case may be available only on Sandbox tiers 2 & 3, but not on tier 1. If you are interested in upgrading your tier access, please contact Awesense.
- For more information please contact tools@awesense.com.